

SUBJECT : Artificial Intelligence

ASSIGNMENT # 02 DUE DATE

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① What is planning in AI? Describe its key components.

Ans. Planning in AI refers to the process of devising a sequence of actions that an agent must take to achieve a specific goal, starting from an initial state, within the constraints of the environment.

* This process ensure that the agent can operate effectively in dynamic or complex scenarios.

Components of AI planning.

① Initial State

* Represents the starting conditions of the environment or system.

② Goal State

* Defines the desired outcomes or conditions that the agent aims to achieve.

③ Actions/operators

* These are building blocks of a plan.

④ State Space

* The set of all possible states that the system can reach by applying actions.

⑤ Plan

* A sequence of actions or a strategy that transitions the system from the initial state to the goal state.

⑥ Constraints

+ conditions that limit the planning process, such as time, resources or rules specific to the domain.

⑦ Planning Algorithms

+ The mechanism used to generate a plan.

⑧ Agent

+ The entity executing the plan.

Q. How is partial order planning different from total order planning? Explain partial order planning with the Blocks world problem.

Ans. Total order planning

+ Here, actions are strictly ordered in a sequence.

+ Each action in the plan has a fixed position relative to others, meaning the order in which actions are executed is fully specified.

+ It generates a single linear sequence of actions that achieves the goal.

+ Eg: A* search or STRIPS planners typically generate total order plans.

Partial order planning

+ Here, actions are not fully ordered but instead only partially constrained based on their dependencies.

+ The order of actions is specified only when necessary to satisfy dependencies.

+ It allows flexibility, enabling multiple valid execution sequences.

+ preferred for complex or dynamic problems as it avoids unnecessary commitments to action order.

Partial order planning in Blocks world problem
+ Block world problem is a classic AI planning problem where the goal is to rearrange blocks on a table to achieve a specific configuration.

Problem Setup:

Initial State:

Goal State

Operators/Actions:

pickUp(x)

putDown(x)

stack(x, y)

unstack(x, y)

Example:

Initial State: Block A is on Block B, Block B is on the table, Block c is on the table.

Goal State: ~~Goal~~ Block c on Block A and Block A on Block B
Stack

③ State the reasons for uncertainty in an environment. Briefly explain different types of uncertainties that may effect problem solving in AI.

Ans. Reasons for Uncertainty:

① Incomplete knowledge: The agent does not have access to all the details about the environment or its state.

② Ambiguity: Information about the environment or actions may be unclear or open to multiple interpretations.

③ Dynamic changes: The environment evolves ~~or~~ unpredictably during the agent's decision-making process.

Types of Uncertainties

1) Uncertainty in State Representation

- * The agent may not know the exact current state of the environment.
- * Eg: A robot navigating a maze with limited sensor data.

2) Uncertainty in perception

- * Sensor data or observations may be noisy, incomplete or distorted.
- * Eg: A camera based system detecting objects in low light conditions.

3) Environmental Uncertainty

- * The external world may change Unpredictable due to factors beyond the agent's control.
- * Eg: Weather variations impacting a drone's flight path.

4) Explain how probabilistic reasoning can deal with uncertainties with an example?

Ans. Probabilistic Reasoning to Handle Uncertainty

- * It deals with uncertainties by using probability theory to represent and infer the likelihood of uncertain events.
- * It helps AI systems make informed decisions even when information is incomplete or noisy.
- * Bayes's Theorem:

$$P(H|E) = \frac{P(E|H)P(H)}{P(E)}$$

- * updates the probability of a hypothesis H based on evidence E .

Example: Diagnosing a Disease

Problem: A doctor wants to know if a patient has Flu (H) based on the symptom fever (E).

- Probability of Flu ($P(H)$): 0.05.
- Probability of fever given flu ($P(E|H)$): 0.8
- Probability of fever overall ($P(E)$): 0.1

Using Bayes's theorem:

$$P(H|E) = \frac{P(E|H)P(H)}{P(E)} = \frac{0.8 \times 0.05}{0.1} = 0.4$$

Result: There is a 40% chance the patient has flu given they have a fever.

5) What is a Markov process? Explain how hidden Markov model can be used to handle uncertainties indicating each of the probabilities.

Ans. Markov process

- * It is a model systems where the next state depends only on the current state, not the past states.
- * It is defined by states and transition probabilities ($P(S_{t+1}|S_t)$).

Hidden Markov Model (HMM)

- * An HMM extends a Markov process by introducing:

1) Hidden States (S): Actual state of the system.

2) Observations: observable signals related to the hidden states.

3) probabilities:

Initial probabilities (π) - Probability of starting in each state: $P(S_0)$.

Transition probabilities (A) - $P(S_{t+1}|S_t)$.

Observation probabilities (B): $P(O_t | I_t)$

Handling Uncertainty With HMM

* It handles uncertainty by inferring hidden states from observation.

* Eg: predict weather based on whether someone carries an umbrella.

• States: Sunny, Raining.

• Observations: Umbrella, No Umbrella.

• Algorithms like Viterbi determine the most likely weather sequence given evidence.

* HMM is widely used in speech recognition, bioinformatics and NLP to deal with uncertainty.

⑥ What is an Expert System? How is it different from a normal software? Describe two of the application areas of expert system.

Ans. Expert System is an AI based computer program designed to simulate the decision-making ability of a human expert in a specific domain.

* It uses a knowledge base of human expertise and an inference engine to solve complex problems, provide advice, etc.

Expert System

* Mimics human expertise to solve complex problems.
* Has a knowledge base containing facts and rules.
* May have the ability to adapt based on new input.

Normal Software

* Performs specific tasks based on predefined instructions.
* Does not rely on a knowledge base.
* Typically does not learn from input.

or knowledge.

* Uses inference rules to simulate expert decision-making.

* Typically designed for a specific domain.

* follows predefined algorithms or instructions.
* can be used in general-purpose applications.

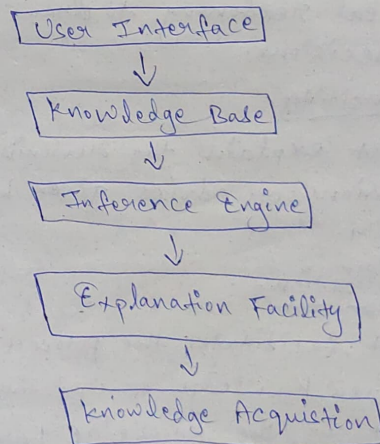
Two application Areas of Expert Systems

1) Medical Diagnosis: Expert Systems are widely used in healthcare for diagnosing medical conditions based on symptoms, patient history and medical tests.

2) Financial advisory System: Expert Systems can assist in making financial decisions, such as investment strategies or credit risk assessments.

⑦ provide a block diagram of an expert system with all its components. Explain the role of each of the components.

Ans. Block Diagram of an Expert System



User interface -

* This is the component that facilitates communication between the user and the expert system.

* It allows users to input data, ask questions and receive solutions or advice.

Knowledge Base -

* It stores all the factual information, rules, heuristics, and domain specific knowledge that the expert system uses to make inference.

* It is the core of the system, containing all the expertise required to solve problems.

Inference Engine -

* It is responsible for applying the rules from the knowledge base to the current problem or data.

* It uses logical reasoning to draw conclusions or make decisions.

Explanation Facility -

* This component explains the reasoning behind the conclusions or advice given by the expert system.

Knowledge Acquisition -

* This component facilitates the process of acquiring new knowledge or updating existing knowledge in the knowledge base.

⑧ Distinguish between forward chaining and backward chaining as the reasoning mechanisms for building an inference engine.

Ans. Forward chaining

* It is a data-driven reasoning method where the system starts with available facts and applies rules to infer new facts or conclusions.

How it works:

- 1) Begin with known facts.
- 2) Apply rules to these facts, if the rule's condition is satisfied, the conclusion becomes a new fact.
- 3) Continue applying rules until a goal is reached or no more applicable rules exist.

Example

In a medical expert system, the rule:
"If fever and cough, then flu"

Characteristics

* Data-driven: starts with facts and moves toward a solution.

* Used when facts are available from the start.

Backward chaining

* It is a goal-driven reasoning method where the system starts with a goal and works backward to find the facts or conditions needed to support the goal.

How it works:

- 1) Start with a goal or hypothesis.
- 2) Find rules that support the goal; check if the conditions are true.

3) work backward to prove or find facts that satisfy these sub-goals.

• Example

To diagnose flu, the system checks if "fever and cough".

Characteristics

- * Goal-driven: Starts with a goal and works backward.
- * Efficient when the goal is known.

⑨ Critically analyze the knowledge engineering process for the development of an effective expert system.

Ans. The Knowledge engineering process is fundamental to building effective expert systems.

* It involves a series of stages aimed at capturing, organizing and applying expert knowledge.

* Each step in the process has its own set of challenges and considerations, which must be carefully addressed to ensure the expert's system's success.

Knowledge Acquisition

- * Gathering knowledge from experts, documents and other sources.
- * Representing complex or ambiguous knowledge is difficult.

Knowledge Representation

* Organizing the acquired knowledge in a

usable format.

* Representing complex or ambiguous knowledge is difficult.

Knowledge Organization

* Categorizing and structuring knowledge for efficient access.

Knowledge Validation and Testing

* Ensuring the system's conclusion are correct through real-world testing.

Knowledge Maintenance

* Regularly updating the system as new information becomes available.

⑩ What is meant by an expert system development life cycle? Explain each phase of the life cycle.

Ans. Expert System Development life cycle

* The ESDLC refers to the structured process followed to design, build, and deploy an expert system.

* It ensures that all necessary components are developed systematically and integrated effectively for the system to function properly.

Phases of the life cycles are:

1) Feasibility Study

* Determine if an expert system is the right solution for the problem.

- * Identify the problem, assess the resources required, and evaluate if the system is feasible given time and budget constraints.

2) Knowledge Acquisition

- * Gather relevant domain-specific knowledge from human experts and other sources.
- * Conduct interviews, review documents, and extract the necessary knowledge to build the knowledge base.

3) Knowledge Representation

- * Organize and represent the acquired knowledge in a formal way for system use.

4) Inference Engine Design

- * Develop the core reasoning mechanism of the system.
- * Design the inference engine that applies the knowledge base to draw conclusions using reasoning techniques like forward or backward chaining.

5) System Testing and Deployment

- * Integrate and test the system before deployment.